

Assignment 19.2

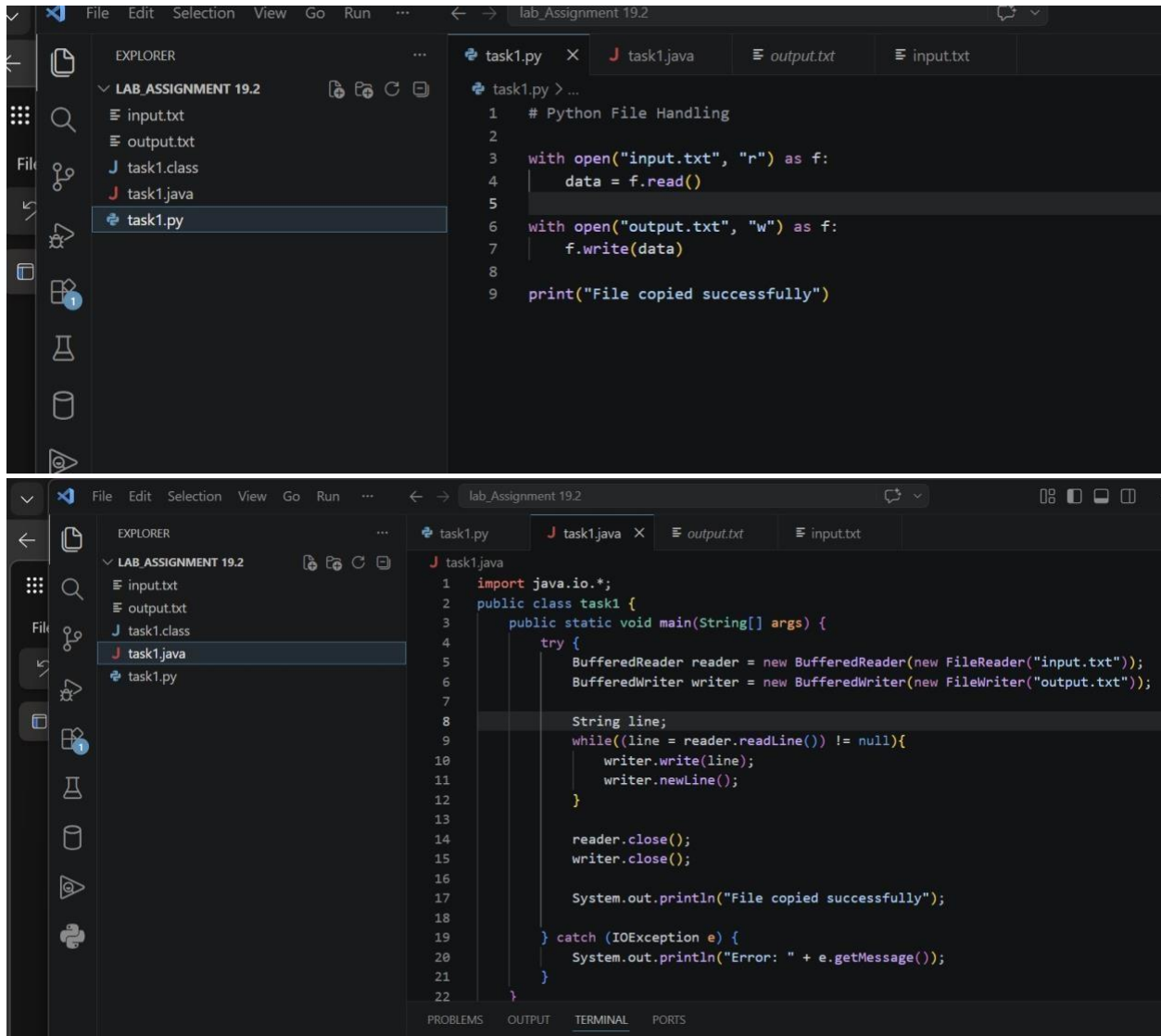
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Task 1: Python to Java Conversion (File Handling) Prompt:

Convert a Python file handling program into Java.

Code:



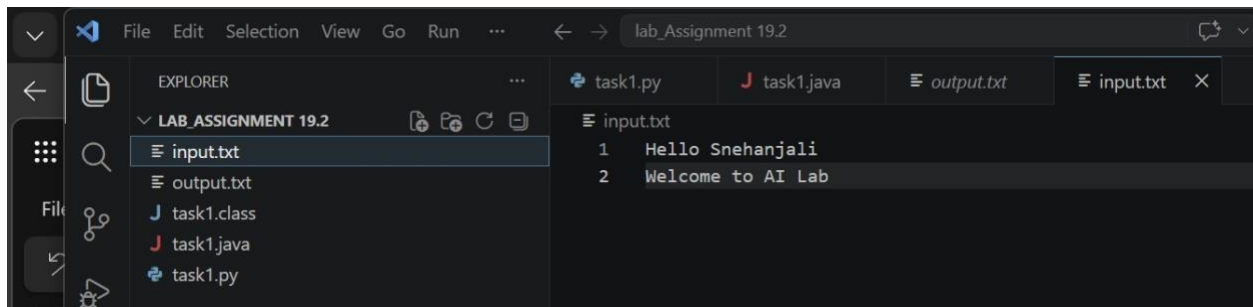
The image displays two screenshots of an IDE (Visual Studio Code) showing the conversion of a Python file handling program into Java.

Top Screenshot: The Python file `task1.py` is open. The code reads from `input.txt` and writes to `output.txt`.

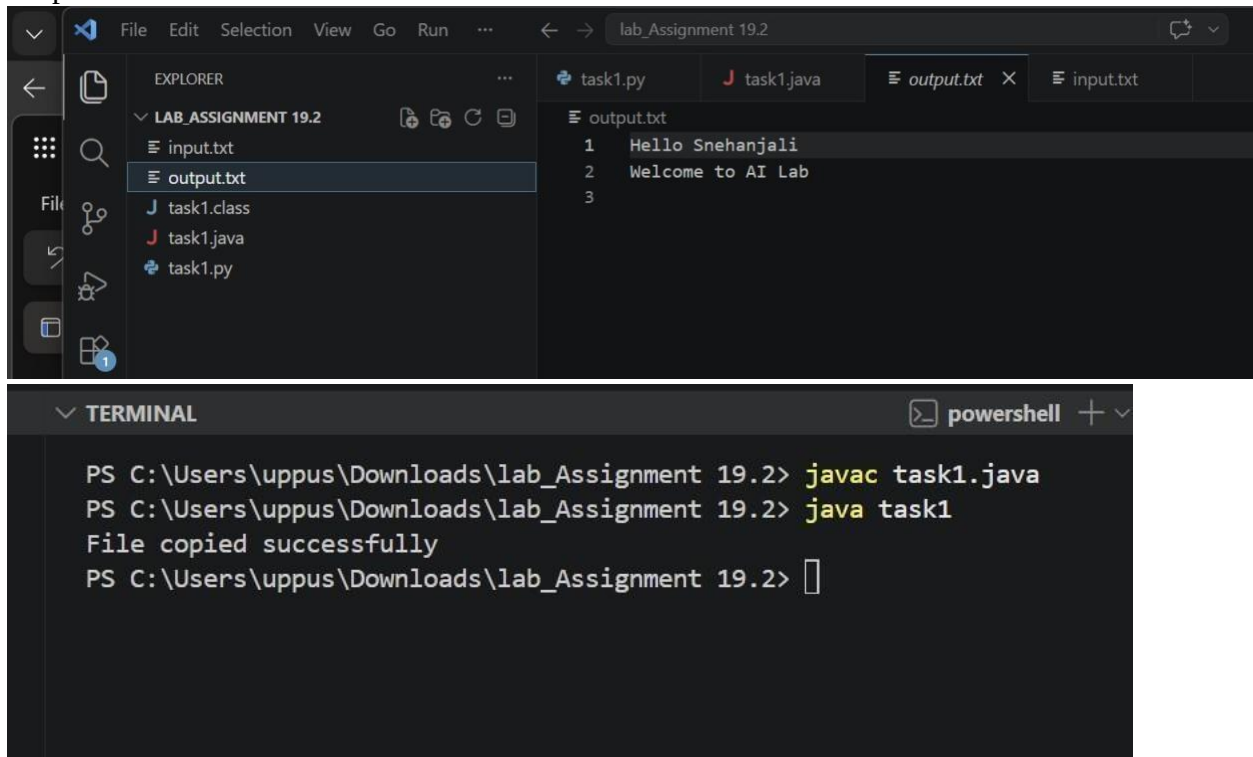
```
1 # Python File Handling
2
3 with open("input.txt", "r") as f:
4     data = f.read()
5
6 with open("output.txt", "w") as f:
7     f.write(data)
8
9 print("File copied successfully")
```

Bottom Screenshot: The Java file `task1.java` is open. The code implements the same functionality using Java's `BufferedReader` and `BufferedWriter`.

```
1 import java.io.*;
2 public class task1 {
3     public static void main(String[] args) {
4         try {
5             BufferedReader reader = new BufferedReader(new FileReader("input.txt"));
6             BufferedWriter writer = new BufferedWriter(new FileWriter("output.txt"));
7
8             String line;
9             while((line = reader.readLine()) != null){
10                 writer.write(line);
11                 writer.newLine();
12             }
13
14             reader.close();
15             writer.close();
16
17             System.out.println("File copied successfully");
18
19         } catch (IOException e) {
20             System.out.println("Error: " + e.getMessage());
21         }
22     }
23 }
```



Output:



Explanation:

The Python program reads data from a file and writes it to another file using open() function.

The Java program uses BufferedReader and BufferedWriter for file handling.

Exception handling is required in Java.

Both programs perform the same operation and produce identical output.

Task2: C++ to Python Conversion (Sorting)

Prompt:

Convert a C++ sorting program into Python.

Code:

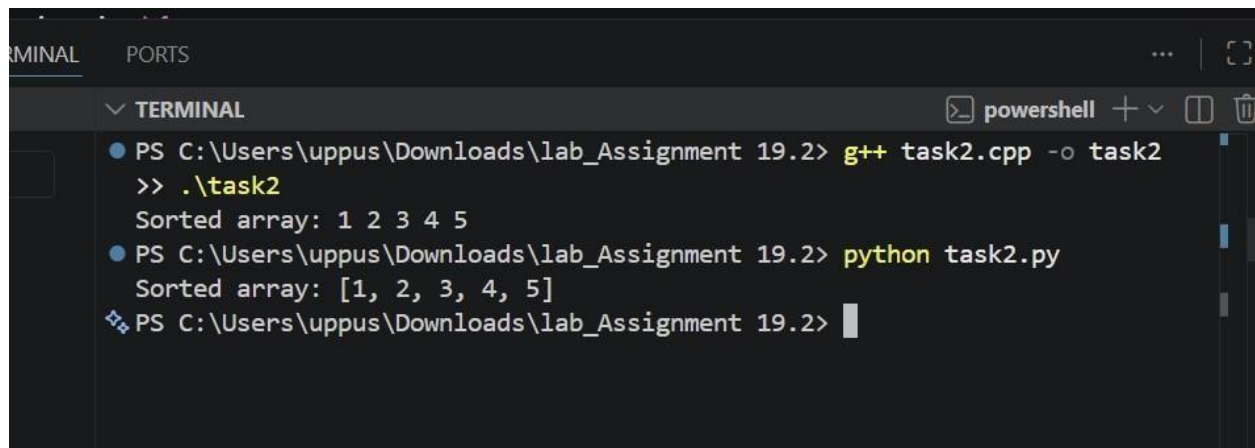
The screenshot shows the Visual Studio Code editor with the 'LAB_ASSIGNMENT 19.2' folder open in the Explorer. The file 'task2.cpp' is selected and open in the editor. The code implements a bubble sort algorithm in C++.

```
task2.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int arr[5] = {5, 2, 4, 1, 3};
6
7      for(int i=0;i<5;i++){
8          for(int j=0;j<5-i-1;j++){
9              if(arr[j] > arr[j+1]){
10                 swap(arr[j], arr[j+1]);
11             }
12         }
13     }
14
15     cout << "Sorted array: ";
16     for(int i=0;i<5;i++)
17         cout << arr[i] << " ";
18
19     return 0;
20 }
```

The screenshot shows the Visual Studio Code editor with the 'LAB_ASSIGNMENT 19.2' folder open in the Explorer. The file 'task2.py' is selected and open in the editor. The code implements a bubble sort algorithm in Python.

```
task2.py > ...
1  arr = [5, 2, 4, 1, 3]
2
3  for i in range(len(arr)):
4      for j in range(len(arr) - i - 1):
5          if arr[j] > arr[j + 1]:
6              temp = arr[j]
7              arr[j] = arr[j + 1]
8              arr[j + 1] = temp
9
10 print("Sorted array:", arr)
```

Output:



```
MINIMAL PORTS
TERMINAL powershell + v
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> g++ task2.cpp -o task2
>> .\task2
Sorted array: 1 2 3 4 5
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task2.py
Sorted array: [1, 2, 3, 4, 5]
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

-->The C++ program uses Bubble Sort to sort an array.

-->The Python program follows the same logic.

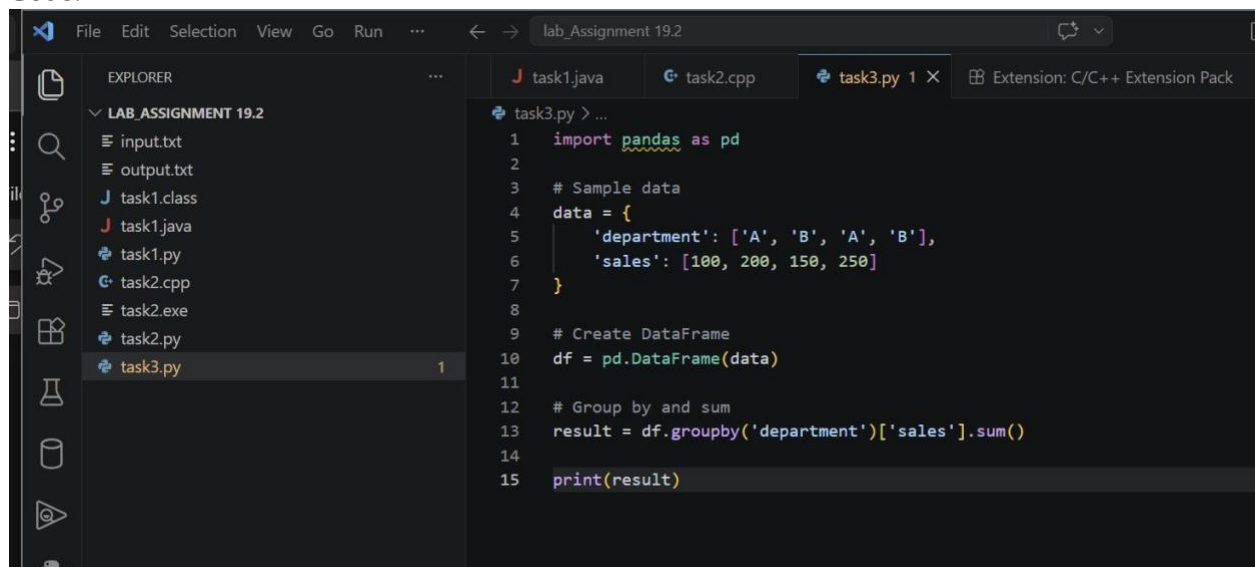
-->Both programs produce the same sorted result.

Task 3: SQL to Pandas Conversion

Prompt:

Convert a SQL query with GROUP BY into Pandas.

Code:



```
File Edit Selection View Go Run ... lab_Assignment 19.2
EXPLORER
LAB_ASSIGNMENT 19.2
input.txt
output.txt
task1.class
task1.java
task1.py
task2.cpp
task2.exe
task2.py
task3.py 1
task3.py > ...
1 import pandas as pd
2
3 # Sample data
4 data = {
5     'department': ['A', 'B', 'A', 'B'],
6     'sales': [100, 200, 150, 250]
7 }
8
9 # Create DataFrame
10 df = pd.DataFrame(data)
11
12 # Group by and sum
13 result = df.groupby('department')['sales'].sum()
14
15 print(result)
```

Output:

```
...
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task3.py
department
A    250
B    450
Name: sales, dtype: int64
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> 
```

Explanation:

-->The SQL query groups data by department and calculates total sales.

-->The Pandas code uses `groupby()` and `sum()` to achieve the same result.

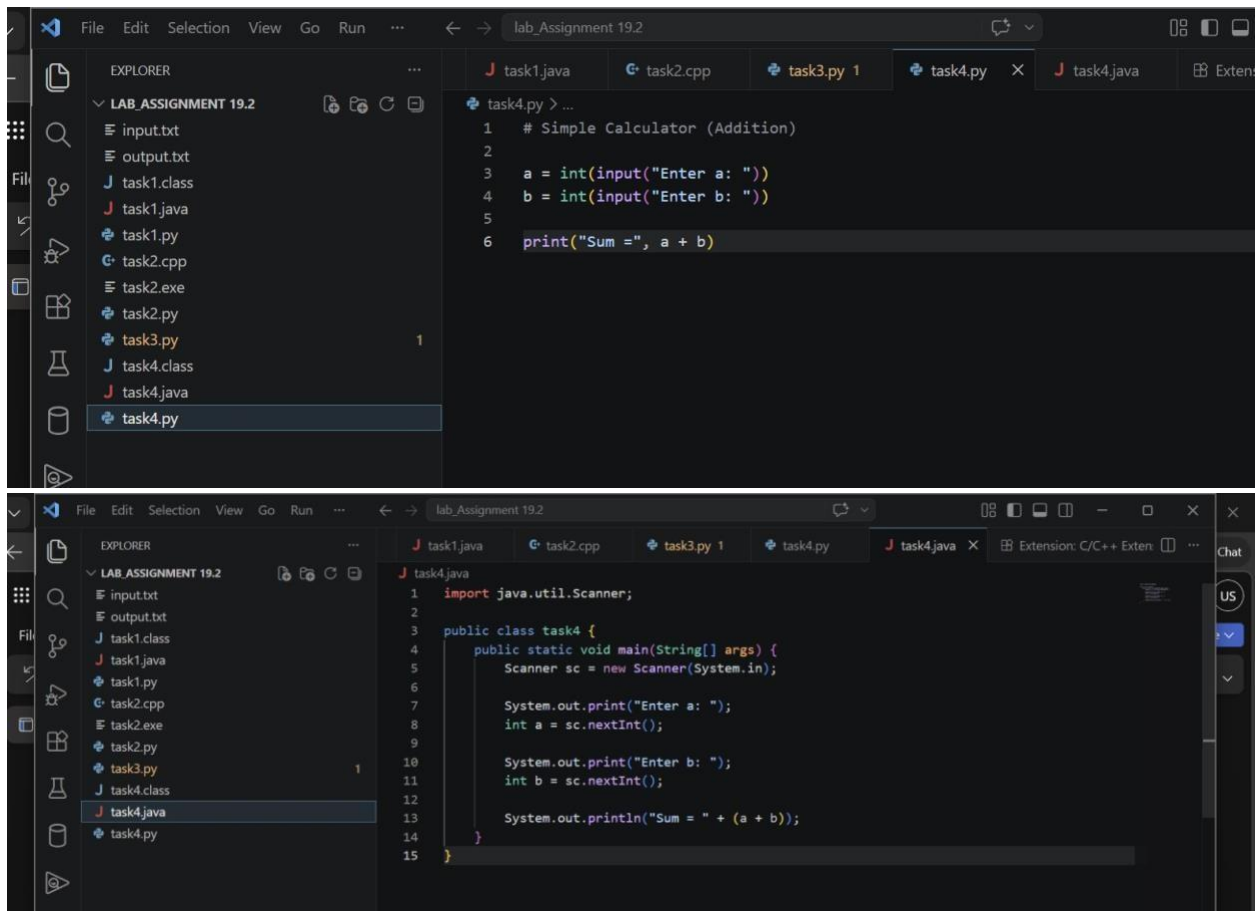
-->Both produce identical outputs.

Task 4: Real-Time Code Conversion (Python to Java)

Prompt:

Convert a Python calculator program into Java.

Code:



Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task4.py
Enter a: 3
Enter b: 5
Sum = 8
```

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> javac task4.java
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> java task4
Enter a: 3
Enter b: 5
Sum = 8
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> 
```

Explanation:

-->The Python program performs addition using simple input and print statements.

-->The Java program uses Scanner for input and follows class-based structure.

-->Both programs perform the same operation and produce identical output.

Task5: Python to C++ Conversion (Even/Odd)

Prompt:

Convert a Python program to C++ to check even or odd number.

Code:

The image shows two screenshots of a code editor interface, likely Visual Studio Code, demonstrating the conversion of a Python program to C++.

Top Screenshot: The editor shows a file named `task5.py` with the following Python code:

```
1 # Check Even or Odd
2
3 n = int(input("Enter number: "))
4
5 if n % 2 == 0:
6     print("Even")
7 else:
8     print("Odd")
```

The Explorer panel on the left shows a project named **LAB_ASSIGNMENT 19.2** with various files including `input.txt`, `output.txt`, `task1.class`, `task1.java`, `task1.py`, `task2.cpp`, `task2.exe`, `task2.py`, `task3.py` (highlighted), `task4.class`, `task4.java`, `task4.py`, `task5.cpp`, `task5.exe`, and `task5.py`.

Bottom Screenshot: The editor shows a file named `task5.cpp` with the following C++ code:

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n;
6     cout << "Enter number: ";
7     cin >> n;
8
9     if(n % 2 == 0)
10         cout << "Even";
11     else
12         cout << "Odd";
13
14     return 0;
15 }
```

The Explorer panel on the left shows the same project structure, but `task5.cpp` is now highlighted.

Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task5.py
Enter number: 3
Odd
● PS C:\Users\uppus\Downloads\lab_Assignment 19.2> g++ task5.cpp -o task5
>> .\task5
Enter number: 3
○ Odd
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> 
```

Explanation:

-->The Python program checks whether a number is even or odd using modulus operator.

-->The C++ program follows the same logic using if-else condition.

-->Both programs produce correct results for all test cases.